

Large CSV File Processing and Validation

This document explains a Python script designed to process large CSV files (2+ GB) efficiently using multiple libraries, validate the data schema, and write the output in a compressed pipe-separated format.

The code reads the file using four different methods (pandas, Dask, Modin, Ray) to compare computational efficiency. It cleans the column names, creates a YAML schema, validates the data against this schema, writes the data in pipe-separated gzipped format, and finally summarizes the file.

Code Explanation by Steps

Step 0: Package Installation

Required Python libraries are installed including dask, modin (with ray backend), ray, and pyyaml for schema handling.

Step 1: Imports and Initialization

Essential libraries are imported, and Ray is initialized to support distributed and parallel processing for Modin and Ray.

Step 2: File Path Setup

The script expects a path to the large CSV file. In Google Colab, this can be an uploaded file or a downloaded file.

Step 3: Define Timing Helper

A utility function measures the time taken by different CSV reading functions to compare efficiency.

Step 4: Reading CSV with Different Libraries

The file is read with pandas (standard), Dask (parallel, chunked), Modin (pandas API with Ray), and Ray Dataset API. Timing is printed for each to assess speed and scalability.

Step 5: Column Name Cleaning

Column names from the pandas dataframe are cleaned by removing special characters and whitespace to standardize schema.

Step 6: YAML Schema Creation

A YAML file is generated that stores the CSV separator and the cleaned list of column names for schema validation.

Step 7: Schema Validation

The script loads the YAML schema and verifies that the ingested dataframe's columns match the expected schema both in count and names.

Step 8: Writing Output File

The cleaned pandas dataframe is written to a pipe-separated ('|') text file compressed with gzip as per the assignment requirements.

Step 9: File Summary

Finally, the script prints a summary including the number of rows, columns, and file size in megabytes for user reference.

Expected Outputs

- **Installation logs for required libraries.**
- **Time taken by each method to read the CSV.**
- **Cleaned column names printed.**
- **Confirmation of YAML schema saved.**
- **Schema validation result messages.**
- **Confirmation of output file written.**
- **Summary of total rows, columns, and file size.**